

Healthy People 2030 (in 2026), Environmental Conditions

Literature Summary & References | Source: odphp.health.gov

About This Literature Summary

This summary of the literature on Environmental Conditions as a social determinant of health is a narrowly defined examination that is not intended to be exhaustive and may not address all dimensions of the issue. Please note: The terminology used in each summary is consistent with the respective references. For additional information on cross-cutting topics, please see the [Quality of Housing](#) literature summary.

Related Objectives (6)

Here's a snapshot of the objectives related to topics covered in this literature summary. [Browse all objectives](#) .

- [Reduce the number of days people are exposed to unhealthy air — EH-01](#)
- [Increase the proportion of people whose water supply meets Safe Drinking Water Act regulations — EH-03](#)
- [Reduce blood lead levels in children aged 1 to 5 years — EH-04](#)
- [Reduce health and environmental risks from hazardous sites — EH-05](#)
- [Reduce the amount of toxic pollutants released into the environment — EH-06](#)
- [Reduce the proportion of adults who have hearing loss due to noise exposure — HOSCD-09](#)

Related Evidence-Based Resources (3)

Here's a snapshot of the evidence-based resources related to topics covered in this literature summary. [Browse all evidence-based resources](#) .

- [Testing for Lead Poisoning in Children](#)
- [Interventions to Prevent Occupational Noise-Induced Hearing Loss](#)



Healthy People 2030 organizes the social determinants of health into 5 domains:

1. Economic Stability
2. Education Access and Quality
3. Health Care Access and Quality
4. Neighborhood and Built Environment

5. Social and Community Context

Literature Summary

Environmental conditions, or the quality and state of the environment, are an inescapable part of daily life that can impact health. Environmental conditions such as water quality, air quality, and weather often vary among populations and geographic locations. While many conditions of the environment are naturally occurring, the quality and impacts of the environment are often affected by historical, economic, and sociopolitical factors.^{1, 2, 3, 4} Certain populations may be more susceptible to poorer environmental conditions and associated health disparities, including communities of color, those with low incomes, those who are unhoused, older adults, pregnant women, and children.^{1, 2, 3, 5} Many aspects of the environment are expected to fluctuate as rising temperatures and shifting weather patterns across the globe cause new and worsening public health challenges, especially among disadvantaged communities.^{2, 5}

Water Quality

Water is an essential part of the environment that is needed for drinking, bathing, and cleaning. Although the United States' water supply is generally safe, water can be contaminated in a variety of ways, such as through certain agricultural practices, sewage leaks, or naturally occurring substances.⁶ Water can be contaminated with pathogens, which cause waterborne diseases, or chemicals, which cause other negative health issues.⁶ There are approximately 7.15 million cases of waterborne illnesses from pathogens alone in the United States each year.⁷

Most people in the United States get their water from public water systems or private wells.⁸ The Safe Drinking Water Act (SDWA) was established in 1974 to regulate drinking water by setting national standards for over 90 contaminants.⁹ However, water quality can be affected by several natural, built, and sociopolitical factors, which can lead to health disparities.¹ For example, location and size of the water source, city planning decisions, and the amount of resources dedicated to managing the water system can all influence a community's water quality.¹ These factors may be more likely to adversely affect certain communities. Several studies have found that communities with lower incomes and communities of color have a higher risk of exposure to water quality health violations and water contaminants, such as nitrates and arsenic.^{1, 10–13} In addition, the SDWA does not regulate small, private wells, which may put some residents at an increased risk for water contamination if the wells are not properly managed.^{14, 15} Rural residents are more likely to use private well water, which can have higher levels of nitrates.¹ High levels of nitrate or arsenic exposure can cause nausea, vomiting, cardiovascular issues, and death.^{16, 17}

Air Quality

Air is another vital condition for life and health. However, air pollution is associated with an estimated 100,000 to 200,000 deaths annually in the United

States. Air quality can be affected when air is contaminated by particle pollution, such as dust, smoke, or drops of liquid, and air pollutants, such as carbon monoxide, ozone, or nitrogen oxides.¹⁸ These particles are commonly emitted into the air by motor vehicles, industrial facilities, and fires.¹⁹ Poor air quality is linked to many health problems, including lung cancer²⁰ and heart disease.²¹, ²² Even naturally occurring particles, such as pollen, can circulate in the air and cause allergic reactions and respiratory issues that substantially impact the public health system.²³, ²⁴

Certain communities are more susceptible to air pollution than others. Urban areas often have more pollution sources, which contributes to worse air quality when compared to rural areas.²⁵ In addition, communities of color encounter air pollution more often than predominantly White communities due to multiple factors.³, ⁴ For example, neighborhoods of color are more likely to be located near pollution sources, such as factories and industrial facilities, as a result of variants in city planning and limited sociopolitical influence of residents.³, ⁴ A national study of 215 United States Census tracts found that Hispanic individuals and non-Hispanic Black individuals were more exposed than non-Hispanic White individuals to airborne particulate matter, such as chlorine, aluminum, and elemental carbon.²⁶

Other Types of Environmental Conditions

Other community factors, such as noise levels and proximity to hazardous waste sites, can also influence health. Noise pollution often refers to the increased exposure of human-made noises, such as traffic, construction, or other mechanical sounds.²⁷, ³¹ Exposure to noise pollution may illicit stress responses and has been associated with hearing loss, among other health issues.²⁷ Some studies found that noise pollution disproportionately affects communities of color and communities with lower incomes.²⁸, ²⁹ In addition, living near hazardous waste sites, such as industrial facilities, municipal waste sites, or incinerators, is also associated with health concerns, including adverse pregnancy outcomes, childhood cancer, and other diseases.³⁰, ³¹ For example, one study found that adults living near a United States coal-burning power plant were more likely to report respiratory, gum, and skin conditions than adults not living near a power plant.³²

Weather

Severe weather can have disastrous effects on health, including direct injury and death, in addition to causing infrastructure damage and environmental degradation. In 2020, weather-related events in the United States caused an estimated 457 fatalities, 1,425 injuries, and \$27.3 million in total damages.³³ In addition, people affected by weather disasters are at an increased risk of long-term effects, including mental health consequences and new or worsening chronic conditions.⁵ Severe weather also puts increased pressure on the health care system by increasing the number of people who require medical care, impeding access to care, and damaging health care infrastructure. For example, flooding may hinder transportation to health care facilities, and power outages may reduce the ability to provide services or close services completely.³⁴

Increases in the frequency and severity of weather events and a shift in weather patterns will exacerbate issues in areas currently experiencing these conditions and create new challenges for areas not accustomed to certain conditions.⁵ In addition to direct health effects from weather disasters, another threat to health will result from reducing certain qualities of the environment. For example, warmer water temperatures and increased flooding facilitate the growth of waterborne pathogens, which increases risk of illness to people exposed through drinking water, fish and shellfish, or recreational water activities.⁵ Similarly, warmer temperatures and droughts can make crops and meat more susceptible to bacteria and toxins, which increases the risk of foodborne illness.⁵ These changes and increasingly severe weather are likely to disproportionately affect people of color and people with lower incomes, often due to location, limited access to resources, and poorer-quality infrastructure.² For example, urban communities that house people of color and people with lower incomes are more likely to have large amounts of heat-trapping surfaces, which puts these individuals at increased risk of severe heat.^{2, 35} In addition, Hispanic or Latino persons are more likely to work jobs that expose them to weather, such as construction and agriculture, which puts them at higher risk for weather-related health consequences and reduced labor hours.³⁶

Strategies for Improvements

Environmental conditions can be improved through policies and programs that support monitoring, regulations, and investing in infrastructure. Federal laws, such as the Clean Air Act and Safe Drinking Water Act, that regulate air and water quality have been effective in reducing the harmful health effects of pollutants.^{37, 38} Environmental conditions can also be improved through programs and initiatives that help communities enhance their infrastructure. For example, the Centers for Disease Control and Prevention's Safe Water for Community Health program provides funding and resources to health departments to improve the quality of private wells. In addition, urban designs that promote increased vegetation on city streets and building roofs can help improve air quality, reduce noise pollution, and relieve excess heat.³⁹ Some strategies and resources can also help residents improve the environmental conditions in their own homes. The Low Income Home Energy Assistance Program is a federal initiative that serves to reduce negative health outcomes from extreme temperatures by helping families cover their energy expenses.⁴⁰ Residents can also use indoor air filters to improve their indoor air quality, which is especially beneficial in areas susceptible to wildfires.⁴¹

While the relationship between health and some environmental conditions is well documented, additional research is needed to comprehensively assess the extent to which environmental conditions affect health and disparities. As environmental conditions continue to evolve, additional innovative and sustainable interventions are needed to address the effects of environmental conditions on health outcomes and disparities.

Citations

1.

Schaider, L. A., Swetschinski, L., Campbell, C., & Rudel, R. A. (2019). Environmental justice and drinking water quality: Are there socioeconomic disparities in nitrate levels in U.S. drinking water? *Environmental Health*, 18 (1), 3. doi: [10.1186/s12940-018-0442-6](https://doi.org/10.1186/s12940-018-0442-6)

2.

Morello-Frosch, R., Pastor, M., Sadd, J., & Shonkoff, S. B. (n.d.). The climate gap: Inequalities in how climate change hurts Americans & how to close the gap. https://dornsife.usc.edu/assets/sites/242/docs/ClimateGapReport_full_report_web.pdf [PDF - 808 KB]

3.

Mikati, I., Benson, A. F., Luben, T. J., Sacks, J. D., & Richmond-Bryant, J. (2018). Disparities in distribution of particulate matter emission sources by race and poverty status. *American Journal of Public Health*, 108 (4), 480–485. doi: [10.2105/AJPH.2017.304297](https://doi.org/10.2105/AJPH.2017.304297)

4.

Woo, B., Kravitz-Wirtz, N., Sass, V., Crowder, K., Teixeira, S., & Takeuchi, D. T. (2019). Residential segregation and racial/ethnic disparities in ambient air pollution. *Race and Social Problems*, 11 (1), 60–67. doi: [10.1007/s12552-018-9254-0](https://doi.org/10.1007/s12552-018-9254-0)

5.

U.S. Global Change Research Program. (2016). The impacts of climate change on human health in the United States: A scientific assessment (pp. 1–312). https://health2016.globalchange.gov/low/ClimateHealth2016_FullReport_small.pdf [PDF - 19.6 MB]

6.

Centers for Disease Control and Prevention. (2024). About Waterborne Disease Surveillance. <https://www.cdc.gov/healthy-water-data/about/>

7.

Centers for Disease Control and Prevention. (2020, December 1). Waterborne disease in the United States. <https://www.cdc.gov/healthywater/surveillance/burden/index.html>

8.

Centers for Disease Control and Prevention. (2024). Drinking Water Sources: An Overview. <https://www.cdc.gov/drinking-water/about/drinking-water-sources-an-overview.html>

9.

United States Environmental Protection Agency. (2015, April 1). Overview of the Safe Drinking Water Act [other policies and guidance]. <https://www.epa.gov/sdwa/overview-safe-drinking-water-act>

10.

Switzer, D., & Teodoro, M. P. (2017). The color of drinking water: Class, race, ethnicity, and Safe Drinking Water Act compliance. *Journal AWWA*, 109 (9), 40–45. doi: [10.5942/jawwa.2017.109.0128](https://doi.org/10.5942/jawwa.2017.109.0128)

11.

Balazs, C., Morello-Frosch, R., Hubbard, A., & Ray, I. (2011). Social disparities in nitrate-contaminated drinking water in California's San Joaquin Valley. *Environmental Health Perspectives*, 119 (9), 1272–1278. doi: [10.1289/ehp.1002878](https://doi.org/10.1289/ehp.1002878)

12.

Balazs, C. L., Morello-Frosch, R., Hubbard, A. E., & Ray, I. (2012). Environmental justice implications of arsenic contamination in California's San Joaquin Valley: A cross-sectional, cluster-design examining exposure and compliance in community drinking water systems. *Environmental Health: A Global Access Science Source*, 11, 84. doi: [10.1186/1476-069X-11-84](https://doi.org/10.1186/1476-069X-11-84)

13.

McDonald, Y. J., & Jones, N. E. (2018). Drinking water violations and environmental justice in the United States, 2011–2015. *American Journal of Public Health*, 108 (10), 1401–1407. doi: [10.2105/AJPH.2018.304621](https://doi.org/10.2105/AJPH.2018.304621)

14.

Lothrop, N., Wilkinson, S. T., Verhougstraete, M., Sugeng, A., Loh, M. M., Klimecki, W., & Beamer, P. I. (2015). Home water treatment habits and effectiveness in a rural Arizona community. *Water* (20734441), 7 (3), 1217–1231. doi: [10.3390/w7031217](https://doi.org/10.3390/w7031217)

15.

Stillo, F., & Gibson, J. M. (2017). Exposure to contaminated drinking water and health disparities in North Carolina. *American Journal of Public Health*, 107 (1), 180–185. doi: [10.2105/AJPH.2016.303482](https://doi.org/10.2105/AJPH.2016.303482)

16.

Agency for Toxic Substances and Disease Registry. (n.d.). ToxFAQs™ for arsenic . Retrieved December 20, 2021, from <https://wwwn.cdc.gov/TSP/ToxFAQs/ToxFAQsDetails.aspx?faqid=19&toxid=3>

17.

Agency for Toxic Substances and Disease Registry. (n.d.). ToxFAQs™ for nitrate and nitrite. Retrieved December 20, 2021, from <https://wwwn.cdc.gov/TSP/ToxFAQs/ToxFAQsDetails.aspx?faqid=1186&toxid=258>

18.

Centers for Disease Control and Prevention. (2024). Air Pollutants . <https://www.cdc.gov/air-quality/pollutants/index.html>

19.

World Health Organization (n.d.). Air pollution . Retrieved December 21, 2021, from <https://www.who.int/westernpacific/health-topics/air-pollution>

20.

Turner, M. C., Krewski, D., Pope, C. A., Chen, Y., Gapstur, S. M., & Thun, M. J. (2011). Long-term ambient fine particulate matter air pollution and lung cancer in a large cohort of never-smokers. *American Journal of Respiratory and Critical Care Medicine*, 184 (12), 1374-1381. doi: [10.1164/rccm.201106-1011OC](https://doi.org/10.1164/rccm.201106-1011OC)

21.

Alexeeff, S. E., Liao, N. S., Liu, X., Van Den Eeden, S. K., & Sidney, S. (2021). Long-term PM_{2.5} exposure and risks of ischemic heart disease and stroke events: Review and meta-analysis. *Journal of the American Heart Association*, 10 (1), e016890. doi: [10.1161/JAHA.120.016890](https://doi.org/10.1161/JAHA.120.016890)

22.

Kaufman, J. D., Adar, S. D., Barr, R. G., Budoff, M., Burke, G. L., Curl, C. L., Daviglus, M. L., Diez Roux, A. V., Gassett, A. J., Jacobs, D. R., Kronmal, R., Larson, T. V., Navas-Acien, A., Olives, C., Sampson, P. D., Sheppard, L., Siscovick, D. S., Stein, J. H., Szpiro, A. A., & Watson, K. E. (2016). Association between air pollution and coronary artery calcification within six metropolitan areas in the USA (the multi-ethnic study of atherosclerosis and air pollution): A longitudinal cohort study. *Lancet*, 388 (10045), 696-704. doi: [10.1016/S0140-6736\(16\)00378-0](https://doi.org/10.1016/S0140-6736(16)00378-0)

23.

Saha, S., Vaidyanathan, A., Lo, F., Brown, C., & Hess, J. J. (2021). Short term physician visits and medication prescriptions for allergic disease associated with seasonal tree, grass, and

weed pollen exposure across the United States. *Environmental Health*, 20 (1), 85. doi: [10.1186/s12940-021-00766-3](https://doi.org/10.1186/s12940-021-00766-3)

24.

Demain, J. G. (2018). Climate change and the impact on respiratory and allergic disease: 2018. *Current Allergy and Asthma Reports*, 18 (4), 22. doi: [10.1007/s11882-018-0777-7](https://doi.org/10.1007/s11882-018-0777-7)

25.

Strosnider, H. (2017). Rural and urban differences in air quality, 2008–2012, and community drinking water quality, 2010–2015—United States. *Morbidity and Mortality Weekly Report, Surveillance Summaries*, 66 . doi: [10.15585/mmwr.ss6613a1](https://doi.org/10.15585/mmwr.ss6613a1)

26.

Bell, M. L., & Ebisu, K. (2012). Environmental inequality in exposures to airborne particulate matter components in the United States. *Environmental Health Perspectives*, 120 (12), 1699–1704. doi: [10.1289/ehp.1205201](https://doi.org/10.1289/ehp.1205201)

27.

Basner, M., Babisch, W., Davis, A., Brink, M., Clark, C., Janssen, S., & Stansfeld, S. (2014). Auditory and non-auditory effects of noise on health. *Lancet*, 383 (9925), 1325–1332. doi: [10.1016/S0140-6736\(13\)61613-X](https://doi.org/10.1016/S0140-6736(13)61613-X)

28.

Casey, J. A., Morello-Frosch, R., Mennitt, D. J., Frstrup, K., Ogburn, E. L., & James, P. (2017). Race/ethnicity, socioeconomic status, residential segregation, and spatial variation in noise exposure in the contiguous United States. *Environmental Health Perspectives*, 125 (7), 1–10. doi: [10.1289/EHP898](https://doi.org/10.1289/EHP898)

29.

Huang, Y.-K., Mitchell, U. A., Conroy, L. M., & Jones, R. M. (2021). Community daytime noise pollution and socioeconomic differences in Chicago, IL. *PLoS ONE*, 16 (8), 1–12. doi: [10.1371/journal.pone.0254762](https://doi.org/10.1371/journal.pone.0254762)

30.

Brender, J. D., Maantay, J. A., & Chakraborty, J. (2011). Residential proximity to environmental hazards and adverse health outcomes. *American Journal of Public Health*, 101 (Suppl 1), S37–S52. doi: [10.2105/AJPH.2011.300183](https://doi.org/10.2105/AJPH.2011.300183)

31.

Orr, M., Bove, F., Kaye, W., & Stone, M. (2002). Elevated birth defects in racial or ethnic minority children of women living near hazardous waste sites. Findings and Direction of the U.S. Superfund Health Research Program, 205 (1-2), 19-27.

32.

Zierold, K. M., Hagemeyer, A. N., & Sears, C. G. (2020). Health symptoms among adults living near a coal-burning power plant. Archives of Environmental & Occupational Health, 75 (5), 289-296. doi: [10.1080/19338244.2019.1633992](https://doi.org/10.1080/19338244.2019.1633992)

33.

National Weather Service. (n.d.). Summary of natural hazard statistics for 2020 in the United States . Retrieved March 2, 2022, from <https://www.weather.gov/media/hazstat/sum20.pdf> [PDF - 113 KB]

34.

Guenther, R., & Balbus, J. (2014, December). Primary protection: Enhancing health care resilience for a changing climate. <https://toolkit.climate.gov/sites/default/files/SCRHCFI%20Best%20Practices%20Report%20final2%202014%20Web.pdf> [PDF - 2 MB]

35.

Hsu, A., Sheriff, G., Chakraborty, T., & Manya, D. (2021). Disproportionate exposure to urban heat island intensity across major U.S. cities. Nature Communications, 12 (1), 2721. doi: [10.1038/s41467-021-22799-5](https://doi.org/10.1038/s41467-021-22799-5)

36.

U.S. Environmental Protection Agency. (2021). Climate change and social vulnerability in the United States: A focus on six impacts . 101.

37.

Nethery, R. C., Mealli, F., Sacks, J. D., & Dominici, F. (2021). Evaluation of the health impacts of the 1990 Clean Air Act Amendments using causal inference and machine learning. Journal of the American Statistical Association, 116 (535), 1128-1139. doi: [10.1080/01621459.2020.1803883](https://doi.org/10.1080/01621459.2020.1803883)

38.

Weinmeyer, R., Norling, A., Kawarski, M., & Higgins, E. (2017). The Safe Drinking Water Act of 1974 and its role in providing access to safe drinking water in the United States. AMA Journal of Ethics, 19 (10), 1018-1026. doi: [10.1001/journalofethics.2017.19.10.hlaw1-1710](https://doi.org/10.1001/journalofethics.2017.19.10.hlaw1-1710)

39.

The City of New York. (n.d.). Cool neighborhoods NYC: A comprehensive approach to keep communities safe in extreme heat. 44.

40.

U.S. Department of Health and Human Services, Office of Community Services (n.d.). Low Income Home Energy Assistance Program (LIHEAP). Retrieved March 4, 2022, from <https://www.acf.hhs.gov/ocs/low-income-home-energy-assistance-program-liheap>

41.

Joseph, G., Schramm, P. J., Vaidyanathan, A., Breysse, P., & Goodwin, B. (n.d.). BRACE technical report series. 36.